# Mock 04

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# 1 INSTRUCTIONS

- **Time:** 120 Minutes.
- **Total Points:** 41.
- **Allowed Materials:** Pen, paper. No calculators.
- **Format:**
  - **Part A:** Multiple Choice (10 questions, 1 point each).
  - **Part B:** Short Open Problems (10 questions, 1.5 points each).
  - **Part C:** Complex Open Problems (8 questions, 2 points each).

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# 2 PART A: Multiple Choice (10 Points)

*Select the single best answer.*

## 2.1 Problem A1

What is the cardinality of the set of all functions $f : \{0, 1\} \to \mathbb{N}$?

(a) Finite (b) $\aleph_0$ (Countable) (c) $\mathfrak{c}$ (Continuum) (d) $2^{\mathfrak{c}}$

## 2.2 Problem A2

Which of the following is the logical negation of the statement: "For every integer $x$, there exists an integer $y$ such that $x + y = 0$"?

(a) For every integer $x$, there exists an integer $y$ such that $x + y \neq 0$.

(b) There exists an integer $x$ such that for all integers $y$, $x + y \neq 0$.

(c) There exists an integer $x$ and an integer $y$ such that $x + y \neq 0$.

(d) For all integers $x$ and $y$, $x + y \neq 0$.

## 2.3 Problem A3

Find the argument (in the principal interval $(-\pi, \pi]$) of the complex number $z = (1 - i)^{10}$.

(a) $-\frac{\pi}{2}$ (b) $\frac{\pi}{2}$ (c) 0 (d) $\pi$

### 2.4 Problem A4

Find the remainder when $14^{2026}$ is divided by 13.

(a) 1 (b) 12 (c) 2 (d) 5

### 2.5 Problem A5

What is the value of the following limit?

$$\lim_{n\to\infty} \frac{5n^3 - 2n + 1}{2n^3 + n^2 - 10}$$

(a) 0 (b) $\frac{5}{2}$ (c) $\infty$ (d) 1

### 2.6 Problem A6

Consider the following Julia code snippet. What is the final value of `s`?

```
S = Set([1, 2, 3])
s = 0
for x in S
    for y in S
        if x != y
            global s += 1
        end
    end
end
println(s)
```

(a) 3 (b) 6 (c) 9 (d) The code produces an error.

### 2.7 Problem A7

Let $A = \{1, 2, 3\}$. What is the size of the symmetric difference $A \triangle \emptyset$?

(a) 0 (b) 1 (c) 3 (d) $\emptyset$

### 2.8 Problem A8

Find the coefficient of $x^3y^2$ in the expansion of $(2x - y)^5$.

(a) $-80$ (b) 80 (c) 40 (d) $-40$

### 2.9 Problem A9

Which of the following conditions ensures that a relation $R$ is a **Partial Order**?

(a) Reflexive, Symmetric, Transitive

(b) Reflexive, Anti-symmetric, Transitive

(c) Irreflexive, Symmetric, Transitive

(d) Reflexive, Symmetric, Anti-symmetric

### 2.10 Problem A10

If $f : A \to B$ is injective and $g : B \to C$ is injective, then the composition $g \circ f$:

(a) Must be surjective.

(b) Must be bijective.

(c) Must be injective.

(d) Is not necessarily injective.

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## 3 PART B: Short Open Problems (15 Points)

*Provide a concise solution. Lengthy derivations are not required.*

### 3.1 Problem B1

Let $f : \mathbb{R} \setminus \{3\} \to \mathbb{R} \setminus \{2\}$ be defined by $f(x) = \frac{2x+1}{x-3}$. Find the inverse function $f^{-1}(y)$.

### 3.2 Problem B2

Sketch the set of points on the complex plane satisfying $|z - 2i| \leqslant |z + 4|$. Provide a geometric description (e.g., "interior of a circle...", "half-plane defined by...").

### 3.3 Problem B3

Find the greatest common divisor of $n^2 + 1$ and $n + 1$ for any $n \in \mathbb{N}$. (Hint: Use the Euclidean algorithm).

### 3.4 Problem B4

Complete the following Julia function to return `true` if a number `n` is a "perfect square" and `false` otherwise.

```
function is_perfect_square(n::Int)
    if n < 0
        return ...................
    end
    root = .......................
    return root * root == n
end
```

### 3.5 Problem B5

Let $A = \{x \in \mathbb{R} : x^2 < 4\}$ and $B = \{x \in \mathbb{R} : x \geq 0\}$. Find $A \setminus B$.

### 3.6 Problem B6

How many permutations of the word "ALGORITHM" contain the substring "MATH"?

### 3.7 Problem B7

Find the supremum and infimum of the set $S = \{2 - \frac{1}{n} : n \in \mathbb{N}\}$.

### 3.8 Problem B8

Prove or disprove with a counterexample: If $A \cup B = A \cup C$, then $B = C$.

### 3.9 Problem B9

Solve the congruence $5x \equiv 2 \pmod 7$. Give the solution in the form $x \equiv a \pmod 7$ where $0 \leq a < 7$.

### 3.10 Problem B10

Construct a polynomial $P(x)$ with integer coefficients of the lowest possible degree such that $1+\sqrt{3}$ is a root.

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## 4 PART C: Complex Open Problems (16 Points)

*Provide rigorous proofs and step-by-step logic.*

### 4.1 Problem C1

Use Mathematical Induction to prove that for all integers $n \geq 1$:

$$\sum_{k=1}^{n} k(k+1) = \frac{n(n+1)(n+2)}{3}$$

### 4.2 Problem C2

Let $\sim$ be a relation on $\mathbb{Z} \times \mathbb{Z}$ defined by $(a, b) \sim (c, d)$ if and only if $a + d = b + c$.

1. Prove that $\sim$ is an equivalence relation.
2. Describe the equivalence class of $(0, 0)$.

### 4.3 Problem C3

Using the $\epsilon - N$ definition of the limit, prove that:

$$\lim_{n\to\infty} \frac{3n}{n+2} = 3$$

### 4.4 Problem C4

Let $A = \{1, 2, \ldots, 2n\}$. We choose $n + 1$ numbers from $A$. Prove (using the Pigeonhole Principle) that there exist two chosen numbers such that one divides the other. (Reference: Classic number theory application, similar to parity arguments in Problem Set 11/13).

### 4.5 Problem C5

Find all complex solutions to $z^4 = -16$. Express your answers in the form $a + bi$.

### 4.6 Problem C6

A "Derangement" is a permutation of $\{1, \ldots, n\}$ where no element appears in its original position. Using the Principle of Inclusion-Exclusion, derive the formula for the number of derangements $D_n$.

### 4.7 Problem C7

Let $f : A \to B$ and $g : B \to C$ be functions. Prove:

1. If $g \circ f$ is injective, then $f$ is injective.
2. If $g \circ f$ is surjective, then $g$ is surjective.

### 4.8 Problem C8

Show that for any integer $n \geq 1$, the number $n^3 + 5n$ is divisible by 6. (Hint: You may use induction or modular arithmetic cases).

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# 5 ANSWER KEY & GRADING RUBRIC

## 5.1 Section A (1 point each)

1. **(b)** $\aleph_0$. The set of functions $\{0,1\} \to \mathbb{N}$ is equivalent to $\mathbb{N} \times \mathbb{N}$, which is countable. (Note: Functions $\mathbb{N} \to \{0,1\}$ would be uncountable).
2. **(b)** $\exists x \forall y (x + y \neq 0)$. Negation of $\forall\exists$ is $\exists\forall\neg$.
3. **(a)** $-\pi/2$. $(1-i) = \sqrt{2}e^{-i\pi/4}$. Raised to 10: $2^5 e^{-i10\pi/4} = 32e^{-i5\pi/2} = 32e^{-i\pi/2}$.
4. **(a)** 1. $14 \equiv 1 \pmod{13}$, so $1^{2026} \equiv 1$.
5. **(b)** 5/2. Highest degree terms dominate.
6. **(b)** 6. The loop iterates $3 \times 3 = 9$ times. $x \neq y$ is true for $9 - 3$ (diagonal) $= 6$ times.
7. **(c)** 3. $A \triangle \emptyset = (A \setminus \emptyset) \cup (\emptyset \setminus A) = A \cup \emptyset = A$. Size is 3.
8. **(b)** 80. Term is $\binom{5}{2}(2x)^3(-y)^2 = 10 \cdot 8x^3 \cdot y^2 = 80x^3y^2$.
9. **(b)** Reflexive, Anti-symmetric, Transitive.
10. **(c)** Must be injective. Composition of injections is injective.

## 5.2 Section B (1.5 points each)

1. **Solution:** $y = \frac{2x+1}{x-3} \implies y(x-3) = 2x+1 \implies yx - 2x = 3y+1 \implies x(y-2) = 3y+1 \implies x = \frac{3y+1}{y-2}$.
   - **Rubric:** 0.5 for swapping variables/setup, 1.0 for correct algebra.
2. **Solution:** Perpendicular bisector of the segment connecting $2i$ and $-4$. The inequality $|z - 2i| \leq |z - (-4)|$ represents the half-plane containing $2i$.
   - **Rubric:** 1.0 for identifying perpendicular bisector, 0.5 for identifying correct side (half-plane).
3. **Solution:** $\gcd(n^2+1, n+1) = \gcd(n^2 - 1 + 2, n+1) = \gcd((n-1)(n+1) + 2, n+1) = \gcd(2, n+1)$. Result is 2 if $n$ is odd, 1 if $n$ is even.
   - **Rubric:** 1.0 for reduction steps, 0.5 for distinguishing even/odd cases.
4. **Solution:** ‘return false‘ (for negative), ‘isqrt(n)‘ or ‘trunc(Int, sqrt(n))‘.
   - **Rubric:** 0.5 for negative check, 1.0 for correct integer root logic.
5. **Solution:** $A = (-2, 2)$, $B = [0, \infty)$. $A \setminus B = (-2, 0)$.
   - **Rubric:** 0.5 for identifying A, 1.0 for correct difference interval.

6. **Solution:** Treat "MATH" as 1 block. Total items: $\{MATH, A, L, G, O, R, I\}$ (remaining letters). Actually letters are A,L,G,O,R,I,T,H,M. MATH uses M,A,T,H. Remaining: L,G,O,R,I. Total 5 distinct + 1 block = 6 items. Permutations = $6! = 720$.

   - **Rubric:** 1.0 for grouping strategy, 0.5 for correct factorial.

7. **Solution:** $S = \{1, 1.5, 1.66, \dots\}$. Increasing sequence approaching 2. Sup = 2, Inf = 1 (at n=1).

   - **Rubric:** 0.75 for Sup, 0.75 for Inf.

8. **Solution:** False. Counterexample: $A = \{1\}, B = \{2\}, C = \{1, 2\}$. $A \cup B = \{1, 2\}$, $A \cup C = \{1, 2\}$, but $B \neq C$.

   - **Rubric:** 0.5 for "False", 1.0 for valid counterexample.

9. **Solution:** $5x \equiv 2 \pmod 7$. Inverse of 5 mod 7 is 3 ($5 \times 3 = 15 \equiv 1$). Multiply by 3: $x \equiv 6 \pmod 7$.

   - **Rubric:** 1.0 for method (inverse or trial), 0.5 for answer.

10. **Solution:** Root $x = 1 + \sqrt{3} \implies x - 1 = \sqrt{3} \implies (x-1)^2 = 3 \implies x^2 - 2x + 1 = 3 \implies x^2 - 2x - 2 = 0$.

    - **Rubric:** 1.5 for correct derivation of polynomial.

### 5.3 Section C (2 points each)

1. **Induction:**

   - Base ($n = 1$): $1(2) = 2$. RHS: $1(2)(3)/3 = 2$. OK.
   - Hypothesis: Assumed true for $k$.
   - Step: $\frac{k(k+1)(k+2)}{3} + (k+1)(k+2) = (k+1)(k+2)[\frac{k}{3} + 1] = (k+1)(k+2)\frac{k+3}{3}$. Matches form for $k + 1$.
   - **Rubric:** 0.5 Base, 0.5 Hypothesis statement, 1.0 Algebraic step.

2. **Equivalence Relation:**

   - Reflexive: $a + b = b + a$.
   - Symmetric: $a + d = b + c \implies c + b = d + a$.
   - Transitive: Add equations $a + d = b + c$ and $c + f = d + e$.
   - Class of (0,0): $\{(a, b) : a + 0 = b + 0\} \implies a = b$. Diagonal line in $\mathbb{Z}^2$.
   - **Rubric:** 1.0 for Proofs (R/S/T), 1.0 for Class description.

3. **Limit:**

   - Draft: $|\frac{3n}{n+2} - 3| < \epsilon$.
   - Algebra: $|\frac{3n-3(n+2)}{n+2}| = |\frac{-6}{n+2}| = \frac{6}{n+2}$.
   - Find N: $\frac{6}{n+2} < \frac{6}{n} < \epsilon \implies n > \frac{6}{\epsilon}$. Let $N > 6/\epsilon$.

- **Rubric:** 1.0 for algebra simplification, 1.0 for N selection logic.

4. **Pigeonhole:**
   - Write every number $x \in A$ as $x = 2^k \cdot m$ where $m$ is odd.
   - The possible odd parts $m$ are $\{1, 3, 5, \ldots, 2n-1\}$. There are exactly $n$ odd numbers.
   - We pick $n+1$ numbers, so by PHP, two numbers must have the same odd part $m$.
   - Let them be $x = 2^j m$ and $y = 2^k m$. One divides the other.
   - **Rubric:** 1.0 for $2^k m$ representation, 1.0 for applying PHP to odd parts.

5. **Complex Roots:**
   - $z^4 = 16e^{i\pi}$.
   - Roots: $z_k = 2e^{i(\pi+2\pi k)/4}$ for $k = 0, 1, 2, 3$.
   - Angles: $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$.
   - Rectangular: $\sqrt{2}(1+i), \sqrt{2}(-1+i), \sqrt{2}(-1-i), \sqrt{2}(1-i)$.
   - **Rubric:** 1.0 for modulus/angles, 1.0 for rectangular forms.

6. **Derangements:**
   - Total permutations $n!$.
   - Property $P_i$: element $i$ is fixed.
   - Use $N - |\cup P_i|$. $|P_i| = (n-1)!$. Intersection of $k$ sets is $(n-k)!$.
   - Result: $D_n = n! \sum_{k=0}^{n} \frac{(-1)^k}{k!}$.
   - **Rubric:** 1.0 for setup of properties, 1.0 for final formula derivation.

7. **Function Proofs:**
   - (1) Injective: Assume $f(x) = f(y)$. Then $g(f(x)) = g(f(y))$. Since $g \circ f$ injective, $x = y$. Proven.
   - (2) Surjective: Let $c \in C$. Since $g \circ f$ surjective, $\exists a \in A$ s.t. $g(f(a)) = c$. Let $b = f(a)$. Then $g(b) = c$. $\exists b$, so $g$ is surjective.
   - **Rubric:** 1.0 for Injective proof, 1.0 for Surjective proof.

8. **Divisibility:**
   - $n^3 + 5n = n^3 - n + 6n$.
   - $6n$ is divisible by 6.
   - $n^3 - n = n(n-1)(n+1)$ is product of 3 consecutive integers. Divisible by 2 (at least one even) and 3 (one multiple of 3). Thus divisible by 6.
   - Sum is divisible by 6.
   - **Rubric:** 1.0 for splitting or factoring, 1.0 for arguing divisibility by 2 and 3.